

Comparison of Manually Derived and Automated Localisations

1 Marking strategy

The two localisation procedures were compared in the same goal-oriented space–time adaptive loop.

Let $\eta_{K,n}$ denote the signed local contribution associated with spatial cell K and time slab I_n . Define

$$s_{K,n} = |\eta_{K,n}|, \quad S = \sum_{n,K} s_{K,n}. \quad (1)$$

The marked set $\mathcal{M}$ was formed by sorting $s_{K,n}$ in descending order and taking the smallest prefix satisfying

$$\sum_{(K,n) \in \mathcal{M}} s_{K,n} \geq \theta S, \quad \theta = 0.2. \quad (2)$$

The spatial refinement set was the union of the marked spatial cells over all time slabs. In addition, time slab I_n was bisected if the fraction of its spatial cells contained in the global marked set was at least $\rho = 0.025$:

$$\frac{|\mathcal{M}_n|}{|\mathcal{T}_n|} \geq \rho. \quad (3)$$

The manually derived run used strong residual plus inequality localisation. The automated run used a cell-bubble reconstruction. Both runs started from 83 spatial degrees of freedom and $N_t = 8$ time slabs, and used the same tolerance and refinement rule.

2 Iteration histories

Table 1 reports the main quantities at every recorded iteration. The global effectivity is the signed ratio $\eta_{\text{global}}/(J - J_h)$. For the automated method, $I_{\text{loc}} = \eta_{\text{local}}/\eta_{\text{global}}$ measures the consistency of the localisation.

Table 1: Iteration histories for the two localisation strategies.

Method	i	N_x	N_t	$ \eta_{\text{global}} $	I_{eff}	Marks
Strong res	0	83	8	9.501×10^{-4}	1.3130	17
Strong res	1	130	9	5.562×10^{-4}	1.0718	100
Strong res	2	293	11	3.380×10^{-4}	1.0436	425
Strong res	3	766	16	3.282×10^{-4}	1.0276	0
Bubble	0	83	8	9.501×10^{-4}	1.3130	30
Bubble	1	150	11	3.529×10^{-4}	1.1038	95
Bubble	2	337	15	2.629×10^{-4}	1.1063	330
Bubble	3	875	20	4.381×10^{-4}	1.0411	0

Here N_x is the number of spatial degrees of freedom, and “Marks” is the number of marked space–time cells before refinement.

Table 2: Localisation efficiency for the automated method.

Iteration i	$I_{\text{loc}} = \eta_{\text{local}}/\eta_{\text{global}}$
0	1.02480
1	1.02582
2	1.01132
3	1.00154